

# Measuring Race at Eight Geographic Scales

What the 2020 census says about race, at eight scales, and why only the smallest one describes anyone

84-355 Democracy's Data

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Eight Mile Road runs east and west along the northern city limit of Detroit. South of it is the city. North of it are the suburbs — Ferndale, Warren, Eastpointe. In 2002 a film called *8 Mile*, starring the rapper Eminem, took its title from that road. It took it because of what the road separates: the mostly Black city on one side, and the largely White suburbs on the other.

In the film the road is not a setting. It is the line the whole story is about being on the wrong side of. That is why a state highway in Michigan is a phrase the rest of the country recognizes without being told where it is.

The census can say how literally true that is. Take every census block within 2.5 km of Detroit's northern boundary, along the whole width of the city. That is 5,445 blocks holding 299,535 people. Sort them by which side of the line they sit on, and how far from it.

In the last 500 metres inside the city, the population is 90.7% Black. In the first 500 meters outside it, 53.5%. **That is a fall of 37.2 points across one road.**

The shape of it matters as much as the size. This is not a city fading into its suburbs. Composition is flat for two and a half kilometers on the south side, steps at the boundary, and is flat again for two and a half kilometers on the north side. Nothing diffuses. Whatever produced this was operating on the line itself, and not on distance from anywhere.

Now the fact that this chapter exists to explain. Along that stretch, Eight Mile is not only the city limit. It is also the county line: Wayne County to the south, Macomb and Oakland to the north. Which means that **a boundary this sharp, and this widely known, does not appear in county data at all.**

Each of those three counties publishes one row of numbers. In that row, its own side of the step has already been averaged together with a great deal of ground far away from it. The line is not concealed and it is not disputed. It is invisible by arithmetic rather than by omission. And it is invisible in exactly the statistics that reporting, funding formulas and most social science reach for first.

This chapter is a zoom. It takes the 2020 census and holds the people and the categories fixed. Then it changes one thing only: the size of the box the average is computed over. The boxes run from the whole country down through regions, states, counties, tracts and block groups, to the blocks along that road. The finding is the one the road already suggests: **the national numbers everyone quotes describe a place that almost no American actually lives in.**

There is a second finding, and it has to come first, because it is about the boxes rather than the map. The categories counted below are not facts about people. They are a fed-

eral administrative decision, rewritten in 1997 and rewritten again in 2024, and the chapter ends by showing what a number does when the category moves and nobody does.

## 01 A file the law required, borrowed for a question it never asked

Every number in this chapter comes out of one file, and that file exists because Congress ordered it. Public Law 94-171, passed in 1975, requires the Census Bureau to hand every state legislature small-area population counts within a year of Census Day. That is so districts can be redrawn on time. The Redistricting Data Summary File is what discharges that duty.

Nobody at the Bureau produced it to answer a question about diversity, or about neighborhoods, or about Eight Mile Road. It was produced so that map-drawers would have blocks to add up. That is exactly why it can be asked the question this chapter asks.

It was fetched on 2026-08-11, with no key needed. It holds the nation, four regions, nine divisions, the states and the District of Columbia, 3,143 counties, 84,414 tracts, 239,780 block groups and 8,132,968 blocks. The same 331,449,281 people, added up 8 different ways.

That is the property nothing else has. A chapter whose only moving part is the size of the box needs every box filled from the same records, and needs the smallest box to be genuinely small. The American Community Survey is the usual alternative, and it stops at the block group. It also attaches a margin of error to every estimate — a stated range for how far off the estimate could be. At those sizes that range is frequently as large as the number it qualifies.

It could not reach the blocks along Eight Mile at all. For this question there is no better file. There is not really a second one.

What the file cannot do it cannot do for anyone. A full count is a design, not an outcome. The Bureau's own follow-up survey concluded that 2020 missed some populations on net and double-counted others. None of that is marked in any row here.

The count of people is also not a count of anything a person is. The eight boxes are a federal decision that has been rewritten twice. That is a subject of this chapter rather than a defect to be cleaned out of it.

And the 2010 file used for the decade comparison holds the same tables in a different physical shape. That makes lining the two up mechanically easy and treacherous by definition.

None of it arrives as a table. "Legacy format" means a geographic header file plus numbered data segments, with no column names anywhere, joined on a record number. The fact that Hispanic origin by race begins at field 77 of segment one is something you learn from a published layout rather than from the file.

The 2010 header is fixed width where the 2020 header is pipe-separated, and its segments are comma-separated where 2020's use pipes. Same fields, same order, two different readers.

Both are read here, with every geographic identifier forced to text so that a state code stays "01" rather than becoming 1. Puerto Rico is dropped on purpose, because it is not in the national total.

The build then refuses to finish unless the eight categories account for the whole population at every level of every state, and each state's blocks sum to the state.

Nor does it arrive in one piece. There is no national block file. The Bureau ships one archive per state, so assembling it meant 51 downloads plus a national roll-up. The compressed archives are enormously larger than the few hundred kilobytes they were boiled down to.

That is tedious rather than hard, and anyone can run it again. No key, no attention needed, and checkable against the same addresses. The boundaries were not even fetched twice. The state outlines and the Michigan tracts are read from two other chapters in this book. That keeps the book from disagreeing with itself, and it also means those chapters' shapes are now this chapter's shapes.

One table on this page came a different way entirely. The three federal directives below were not downloaded from anywhere. Their dates, citations and category counts were read off the Federal Register by a person and typed in by hand. A rule published as prose in 1977 has no machine-readable version and never will.

It appears here as a tidy table, and that is precisely the risk. Nothing about it distinguishes a typed date from a computed one. It cannot be fetched again or compared against anything, and if a digit in it is wrong, the only way to find out is to go back to the *Register* and read.

## 02 The line the whole chapter is a reading of

"Legacy format" is easier to see than to describe. Here are eight rows of the national roll-up exactly as the Bureau ships them: four from the geographic header, four from the segment that carries the counts. Every number in the rest of this section is computed out of them in front of you.

Here is the United States, in the geographic header file:

| Field | Value         | What it means                              |
|-------|---------------|--------------------------------------------|
| 1     | PLUS          | —                                          |
| 2     | US            | —                                          |
| 3     | 010           | summary level — 010 means the whole nation |
| 4     | 00            | —                                          |
| 5     | 00            | —                                          |
| 6     | 000           | —                                          |
| 7     | 00            | —                                          |
| 8     | 0000001       | logical record number — the join key       |
| 9     | 0100000US     | —                                          |
| 85    | 9150526749046 | —                                          |
| 86    | 698732674530  | —                                          |
| 87    | United States | —                                          |
| 88    | United States | the name of the thing this row describes   |
| 89    | A             | —                                          |

| Field        | Value        | What it means                       |
|--------------|--------------|-------------------------------------|
| 91           | 331449281    | —                                   |
| 92           | 140498736    | —                                   |
| 93           | 38.61252775  | —                                   |
| 94           | -92.39729495 | —                                   |
| 95           | 00           | —                                   |
| the other 77 | (empty)      | kinds of geography a country is not |

Those empty fields are not damage. There are 78 of them, one for each kind of geography a country is not: no county code, no tract, no school district. The row is 97 fields wide because every row in the file is, whether it is the United States or a census block in Kansas. Field 3, 010, is the **summary level**. It is the only thing distinguishing this row from the 3,143 county rows, the 84,414 tract rows, and the region and division rows that share the file. Here are three more, chosen one per level, with the fields that separate them pulled out of rows that are otherwise identical in shape:

| Summary level | What that code means | Logical record | Name in field 88            |
|---------------|----------------------|----------------|-----------------------------|
| 010           | the whole nation     | 0000001        | United States               |
| 020           | a region             | 0000002        | Northeast Region            |
| 030           | a division           | 0000008        | East North Central Division |
| 040           | a state              | 0000031        | Kansas                      |

010, 020, 030, 040: nation, region, division, state. **Four different kinds of thing, in one file, told apart by a three-digit code that appears nowhere in the row's own name.**

Now the same four records in the file that holds the counts. This is the United States, all 149 fields of it:

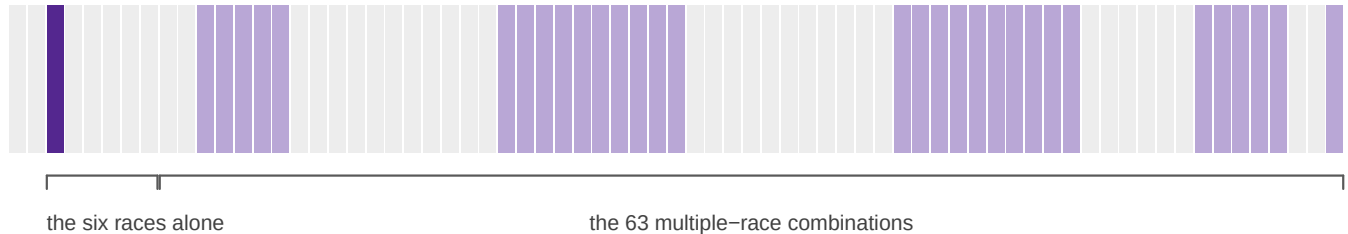
| Field  | Value     | What it means                        |
|--------|-----------|--------------------------------------|
| 1      | PLUS      | file identifier                      |
| 2      | US        | state abbreviation                   |
| 3      | 000       | characteristic iteration             |
| 4      | 01        | file sequence number                 |
| 5      | 0000001   | logical record number — the join key |
| 6      | 331449281 | population of the United States      |
| 7      | 297600338 | population of one race               |
| 8      | 204277273 | White alone                          |
| 9-77   | counts    | the rest of Table P1                 |
| 78     | 62080044  | Hispanic population                  |
| 79-149 | counts    | Table P2                             |

Five label fields, then 144 numbers with nothing attached to them. Field 6 is 331,449,281, the population of the United States. Field 78 is 62,080,044, the Hispanic population. Field

8 is 204,277,273, the number of people who marked White and nothing else. **The file says none of that.** The layout is published as a separate document, and a reader without it has a line of digits.

Two of this chapter's headline numbers are in that line and one of them is not in it anywhere. 204,277,273 people marked White alone. But 235,411,507 marked White *at all*, and no field holds that figure: it is field 8 plus every one of the multiple-race fields that includes White, which is 31 more of them.

field 1: the total



**Where “White alone or in combination” comes from.** The 71 fields of Table P1 for one row of the file. The dark cell is “White alone”. The 31 mid-tone cells are the multiple-race combinations that include White; adding all 32 together is the only way to get the second number, and nothing in the file marks which cells to add.

Everything above is arithmetic anyone can check, and this chapter checks it: the chunk that drew that figure refuses to run unless adding those 32 fields out of the captured line reproduces 235,411,507 exactly.

And here is the row this chapter reads, alongside the position each number came from:

| Column      | Value       | Where it came from                 |
|-------------|-------------|------------------------------------|
| total       | 331,449,281 | segment 1, field 6                 |
| hispanic    | 62,080,044  | segment 1, field 78                |
| nh_white    | 191,697,647 | segment 1, field 81                |
| nh_two      | 13,548,983  | segment 1, field 87                |
| white_alone | 204,277,273 | segment 1, field 8                 |
| white_any   | 235,411,507 | 32 fields of segment 1, added      |
| entropy     | 1.256       | computed from the eight P2 shares  |
| geoid       | (empty)     | the United States has no FIPS code |

Six decisions stand between the pipes and that table.

**A code decides what kind of thing a row is.** 010 is the nation and 040 is a state, and the build selects on those codes. The national roll-up holds 2,832 records across 13 summary levels; this chapter uses four of them and reads the rest of its ladder out of 51 separate state archives.

**Two files are joined on a number that means nothing outside them.** The geographic header carries 0000001 in field 8; segment 1 carries 0000001 in field 5. That is the only thing tying the counts to the place. It is unique inside this archive and meaningless outside it, which is why the join happens before anything else does.

**The United States has an empty identifier, and the emptiness is a fact.** The identifier cell for the nation is blank because there is no code for the country. Not because a value went missing.

**A blank meaning “no such thing” and a blank meaning “we do not know” look identical in every table ever printed.** Only the person who wrote the file can tell you which one this is.

**Two ways of being White are both published, and both are kept.** 204,277,273 alone; 235,411,507 alone or in any combination. Neither is the right one. The census-race lab next door reads this same file and keeps only the first, because its question is about the two-question form rather than about multiple race. **Same line, two chapters, different columns – and the file did not choose.**

**The \_any columns are summed by generated index, not typed.** Getting `white_any` means knowing the order of the 63 combination fields. Two-race combinations come first, then three, then four, five, six, each in alphabetical order over the six races. That index is generated rather than typed. Typing 31 field numbers by hand is how exactly one of them comes out wrong and nothing complains.

The check in this document is the same idea pointed backwards.

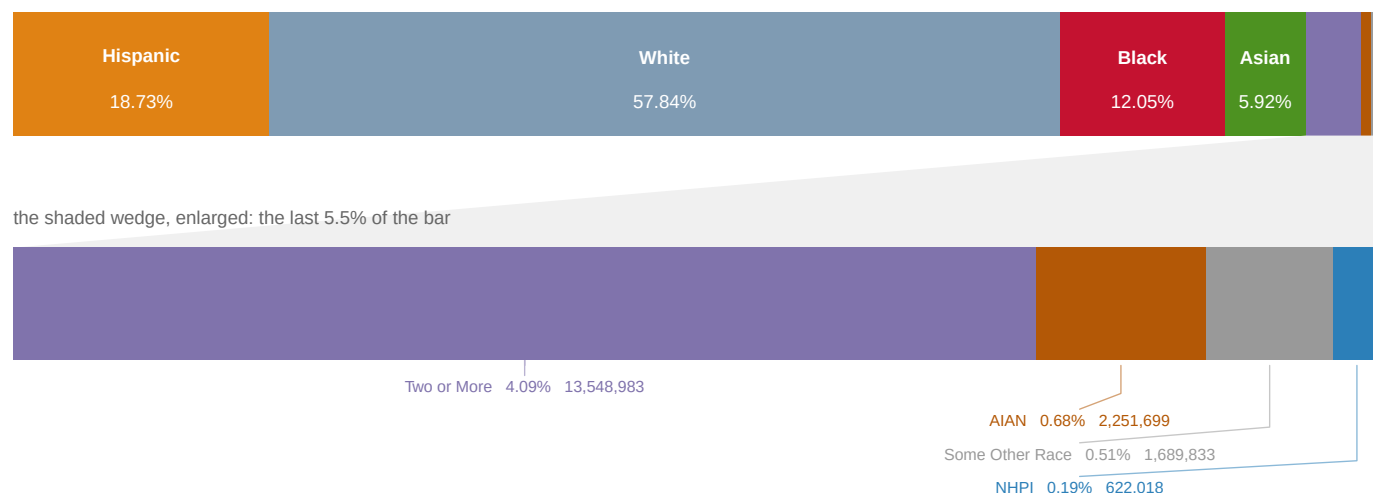
**The diversity measure is computed, and computed over the eight categories rather than the sixty-three.** `entropy` is in no census file. It is calculated here from the eight non-overlapping categories, and it has to be those eight. A measure of how a population divides needs categories that every person falls into exactly once, and the alone-or-in-combination counts do not have that property. **The choice of which published table to measure over is the whole measurement,** and this chapter comes back to it at every rung of the ladder.

## 03 What the census counted

Start with the count itself. On 1 April 2020 the census recorded 331,449,281 people in the fifty states and the District of Columbia, and sorted every one of them into exactly one of eight categories.

every resident of the United States, one category each

2020 Census, Table P2 – total population – 331,449,281 people



**Figure 1. The United States in 2020, as the census sorted it.** Eight mutually exclusive categories, in true proportion. Because five of them are under six percent, the last 5.5%

of the bar is drawn again beneath it, enlarged. 57.8% of the country is non-Hispanic White, 18.7% is Hispanic or Latino, 12.1% is non-Hispanic Black. The smallest category, non-Hispanic Native Hawaiian and Other Pacific Islander, is 622,018 people, or 0.19%.

That table is the one most reporting uses, and it has a property worth naming before anything else. **Every person falls into exactly one of its eight rows, and nobody falls into none.**

Hispanic origin is asked as a separate question from race, so the Bureau has to decide what to do with the millions of people who answer both. The arrangement above settles it by counting Hispanic respondents once, as Hispanic, whatever they said about race. That decision is the subject of the *census-race* chapter, which takes apart Questions 8 and 9 of the 2020 form, the write-in boxes, and the strange career of “Some Other Race.” Nothing here repeats it.

What matters here is only this. A decision was made, a different one was available, and the eight rows above are its output rather than a fact about people.

## 04 Nobody at the Census Bureau chose these boxes

The categories are not the Bureau’s. They are set by the **Office of Management and Budget**, in a document called Statistical Policy Directive No. 15. They bind every federal agency that collects data on race: the Bureau of Labor Statistics, the Department of Education, the FBI’s crime reporting, the census. The Bureau’s job is to implement them.

The directive has been issued three times.

| Issued      | What                 | Categories | Question form                    | First census   |
|-------------|----------------------|------------|----------------------------------|----------------|
| 12 May 1977 | Directive No. 15     | 5          | separate collection preferred    | 1980           |
| 30 Oct 1997 | Revised, 62 FR 58782 | 6          | two separate questions preferred | 2000           |
| 29 Mar 2024 | Revised, 89 FR 22182 | 7          | one combined question required   | 2030 (planned) |

Three provisions of those three documents do most of the work in this chapter.

**1977 allowed one mark.** Somebody of mixed ancestry was to be reported, in the directive’s words, in “the category which most closely reflects the individual’s recognition in his community.”

There was no arithmetic for a person of two races, because there was no such person in the data.

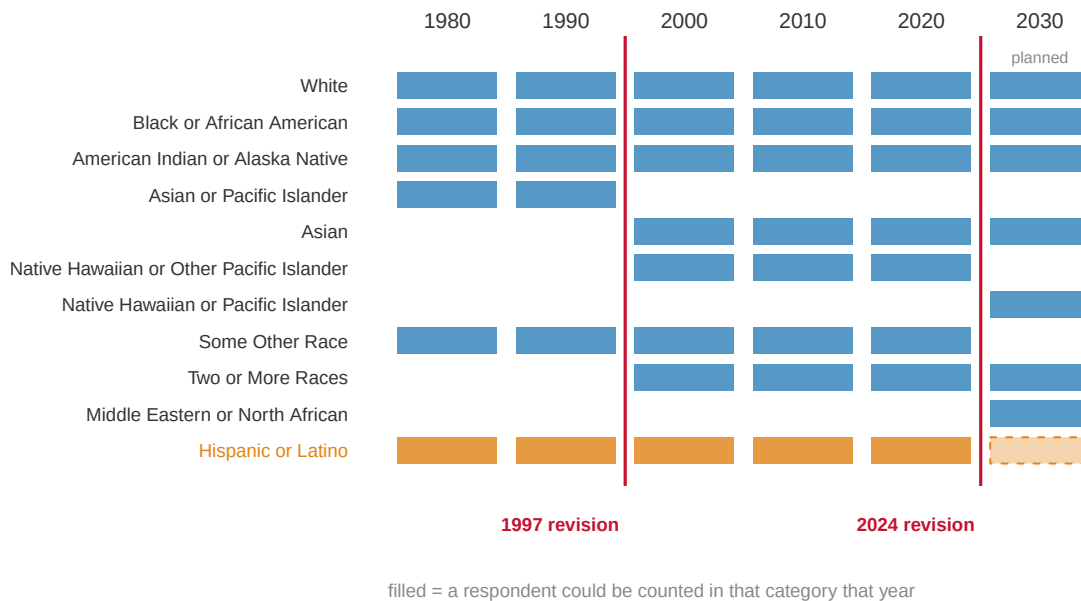
**1997 required more than one.** The revision published on 30 October 1997 (62 FR 58782) says that people “shall be offered the option of selecting one or more racial designations.”

It also split the single 1977 category “Asian or Pacific Islander” into two. The first census to run under it was 2000.

**2024 collapsed the two questions into one.** The revision of 29 March 2024 (89 FR 22182) requires “a combined race and ethnicity question.” It moves Hispanic or Latino into the same list as the races. And it adds a new minimum category: **Middle Eastern or North African.**

There are now 7 minimum categories rather than 6.

Figure 2 draws all of it: which box a person could actually be counted in, in each census since 1980, with the two revisions marked where they fall.



**Figure 2. Which box you could be counted in, by census year.** Each filled cell is a category somebody could actually be recorded as in that census. The red rules are the two revisions of the federal standard. Every appearance and every disappearance in the figure sits against one of them. “Two or More Races” does not exist before 2000, because until the 1997 revision the form did not allow a second mark.

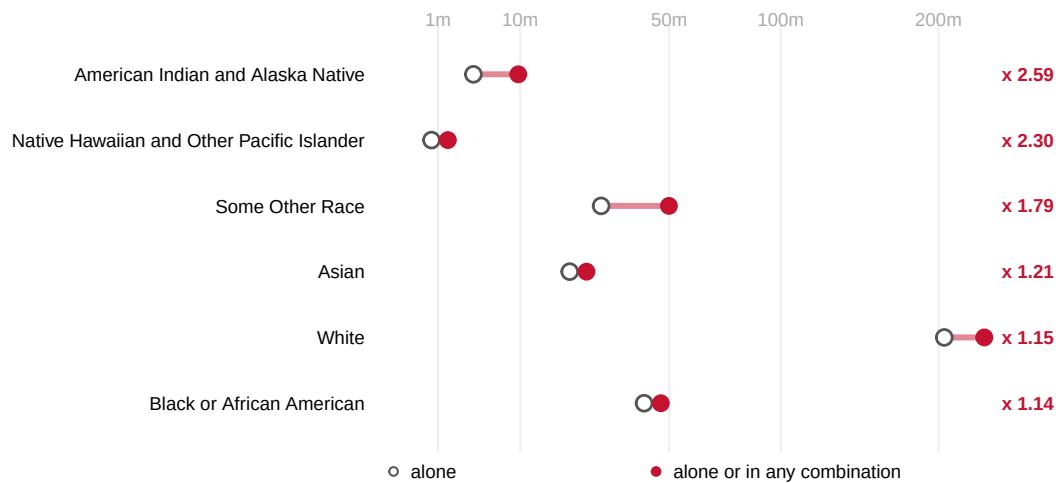
“Middle Eastern or North African” appears only in the last column, planned for 2030. Before that, the 1977 directive defined White as covering “origins in any of the original peoples of Europe, North Africa, or the Middle East,” so the category absorbed them. Orange is the separate ethnicity question, and the dashed cell in the final column is that question being folded into the race list.

Two things follow, and the second is the one that matters for the rest of this chapter.

The first is that **a category is not a discovery**. “Two or More Races” was not found in the population in 2000. It was created in 1997 by a sentence in the Federal Register, and the eleven million people it described in 2000 had been in the country all along, in other boxes.

The second is that **once a person can mark two boxes, every count has two correct versions**. A man who marks White and American Indian is one person. He appears once in the count of people who are White *alone or in any combination*, and not at all in the count of people who are White *alone*. Both counts are published. Neither is a correction of the other.

**Before you read on, commit to a number.** Below, each of the six race categories is counted both ways: alone, and alone-or-in-combination. **Which category grows the most, in proportion, when you allow the second mark — and by what factor?** Write down the name and a multiplier. Is the largest one half again as big, twice as big, five times as big?



**Figure 3. The same six races, counted twice.** The open circle is the count of people who marked that race and no other; the filled circle is everyone who marked it at all. The multiplier on the right is the ratio between them. **American Indian and Alaska Native goes from 3,727,135 to 9,666,058, a factor of 2.59.**

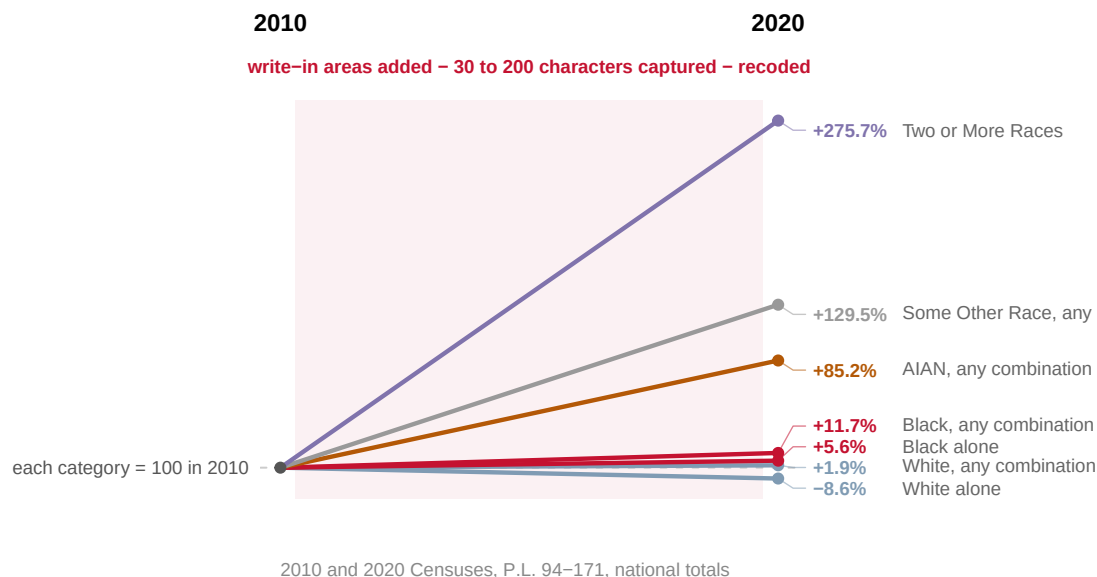
Native Hawaiian and Other Pacific Islander nearly doubles, at 2.30. White moves by the largest absolute amount, 31,134,234 people, and by the smallest proportion, 1.15. Horizontal position is on a square-root scale so the smaller categories remain visible beside White.

The answer people give is Black or White, because those are the categories a reader has been thinking about for two pages. It is neither.

**The category that nearly triples is American Indian and Alaska Native.** The reason is not mysterious once seen. A group of 3,727,135 people has far more of its identifying population married into and descended from other groups, proportionally, than a group of two hundred million does. The smaller the category, the more the second mark matters. So the choice between “alone” and “alone or in combination” moves the small categories most. And the small categories are exactly the ones that voting-rights law, tribal funding formulas and civil-rights enforcement are written about.

## 05 A decade that mostly did not happen

Now put the two censuses side by side. The second mark is not only a fact about a form; it is the reason the most widely reported demographic number of the last decade is unreadable.



**Figure 4. Seven counts of the same country, ten years apart, each set to 100 in 2010.**

The shaded band is not empty space. Between these two censuses the Bureau changed how it processed answers. It added write-in areas under “White” and under “Black or African American.” It raised the characters captured from each write-in from 30 to 200. And it began assigning up to six detailed codes per answer instead of narrowing to two.

**White alone falls 8.6% and White in any combination rises 1.9%.** Two or More Races rises 275.7% and Some Other Race in combination rises 129.5%. Any line drawn across this band measures the population and the processing at once. The figure cannot separate them.

The Census Bureau says so itself, and it is worth quoting exactly, because it is unusually blunt for an agency describing its own product. Of the jump in Some Other Race the Bureau’s own analysts wrote that the change “could be attributed to a number of factors, including demographic change since 2010. But we expect they were largely due to the improvements to the design of the two separate questions for race and ethnicity, data processing and coding.”

**Largely due to the improvements.** Not to migration, not to births, not to anybody changing their mind about who they are. The country did change over the decade. It grew by 22,703,743 people, and the Hispanic share went from 16.3% to 18.7%. But the headline numbers of that decade, the ones about a declining White majority and an exploding multiracial population, are partly an artifact of a better form.

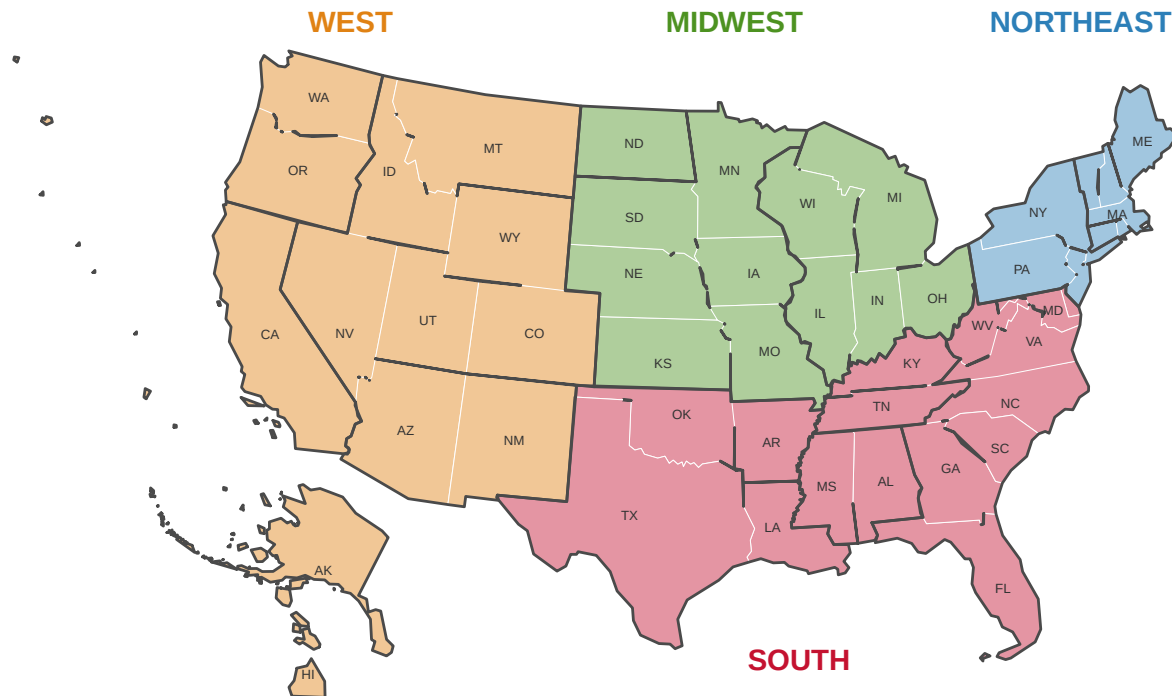
So **this data will not support a clean time series of race.** The honest thing is to say so rather than draw one. That is the whole cross-decade lesson, and it is worth more than the trend line would have been.

Note what this is *not*. 2010 and 2020 were the easy pair. Both ran under the same 1997 standard, with the same categories, in the same file format, with the same fields in the same order. Everything that could have been held constant was held constant, and the

comparison still fails. A comparison that crosses 1997, or that will cross the 2024 revision, has no chance at all.

## 06 Where people are

Turn from time to place, where the data is far better behaved. Composition varies enormously across the country, and the four census regions are the coarsest way to see it. Figure 5 draws them: every state in exactly one division, every division in exactly one region.

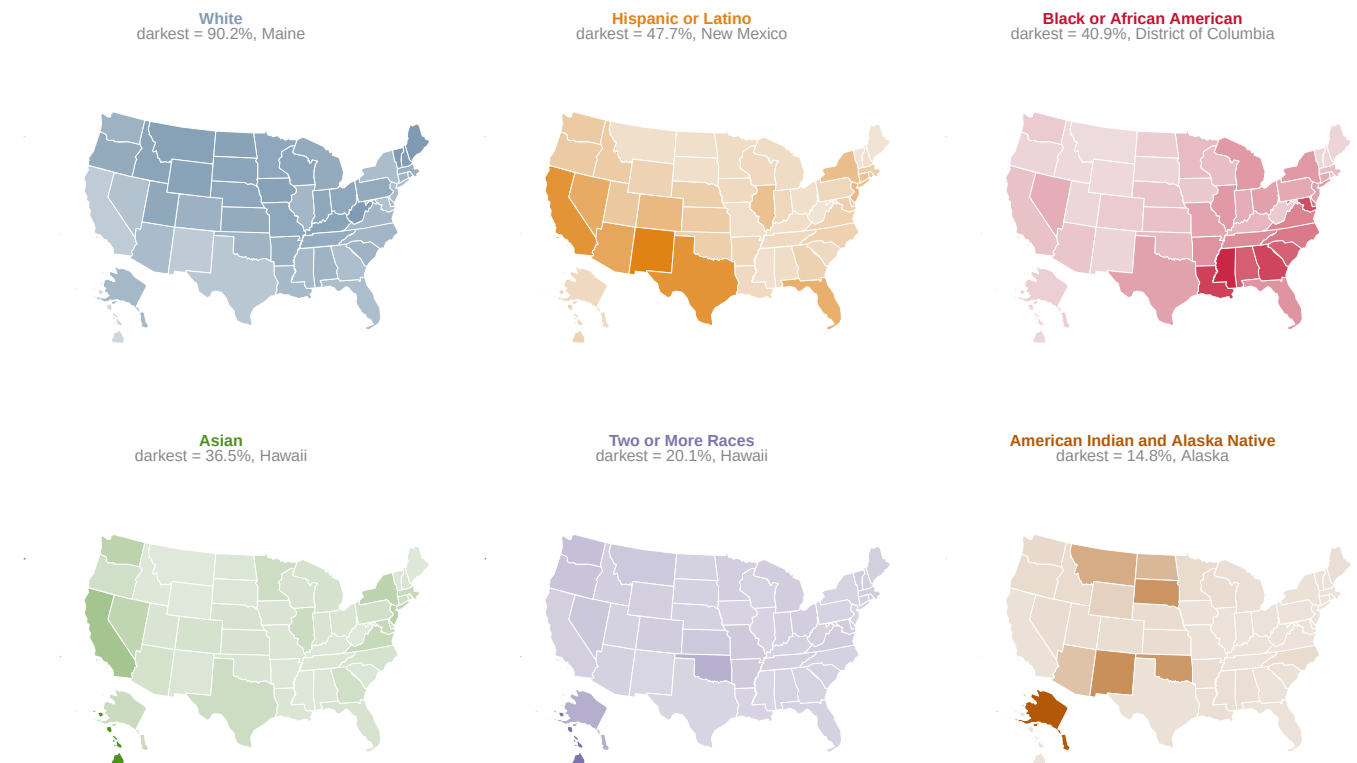


**Figure 5. The four census regions and nine divisions.** The Bureau's own grouping of states, from the same file as every count in this chapter: summary level 020 is a region, 030 a division, and each is exactly a union of states. The heavy lines are the division boundaries. Alaska and Hawaii, both in the Pacific division, are drawn inset at reduced scale; the District of Columbia, in the South Atlantic division, is on the map but too small to label. The outlines and projection are the ones Figure 6 reuses, and the four hues are the ones `regional-shift` gives the same four regions.

| Region    | Population  | Hispanic | White | Black | Asian | Diversity score |
|-----------|-------------|----------|-------|-------|-------|-----------------|
| South     | 126,266,107 | 18.8%    | 54.2% | 18.3% | 3.8%  | 60.9            |
| West      | 78,588,572  | 30.0%    | 47.7% | 4.4%  | 10.6% | 65.0            |
| Midwest   | 68,985,454  | 8.7%     | 72.6% | 10.3% | 3.5%  | 47.0            |
| Northeast | 57,609,148  | 15.3%    | 62.1% | 10.8% | 7.2%  | 57.1            |

The West is the most mixed of the four and the Midwest the least, and the gap between them is wide. But regions are enormous. The smallest holds 57,609,148 people, and aver-

aging at that size destroys almost everything. States are better, and Figure 6 maps all six of the largest categories across all fifty-one of them.



**Figure 6. Six categories, fifty-one states, one projection.** Each panel is scaled to its own maximum, because the categories differ enormously in size. The darkest state and its value are printed under each title. The patterns are the familiar ones, and they are geographic rather than random.

Hispanic population runs along the southern border and into the industrial Northeast. Black population runs through the South Atlantic and East South Central divisions. Asian population sits on the Pacific coast and around Washington and New York.

American Indian and Alaska Native concentrate in Alaska, Oklahoma and the Mountain division.

Alaska and Hawaii are moved and rescaled in the conventional way.

State-level composition ranges from 22.8 on this chapter's diversity score in Maine to 75.6 in Hawaii. Only 4 of the fifty-one have a largest group that is not White: California, District of Columbia, Hawaii, New Mexico.

## 07 The number that keeps moving

Here is where the chapter turns. Everything so far has described containers: the country, four regions, fifty-one states. Each container has been reported by its average. But an average over a container the size of Texas is not a description of anywhere. So keep the population absolutely fixed, keep the categories absolutely fixed, and change only the size of the box you compute the average over.

To make that produce a number rather than a series of prettier maps, this chapter uses one measure at every rung: the **multigroup entropy index**. For any unit, entropy is

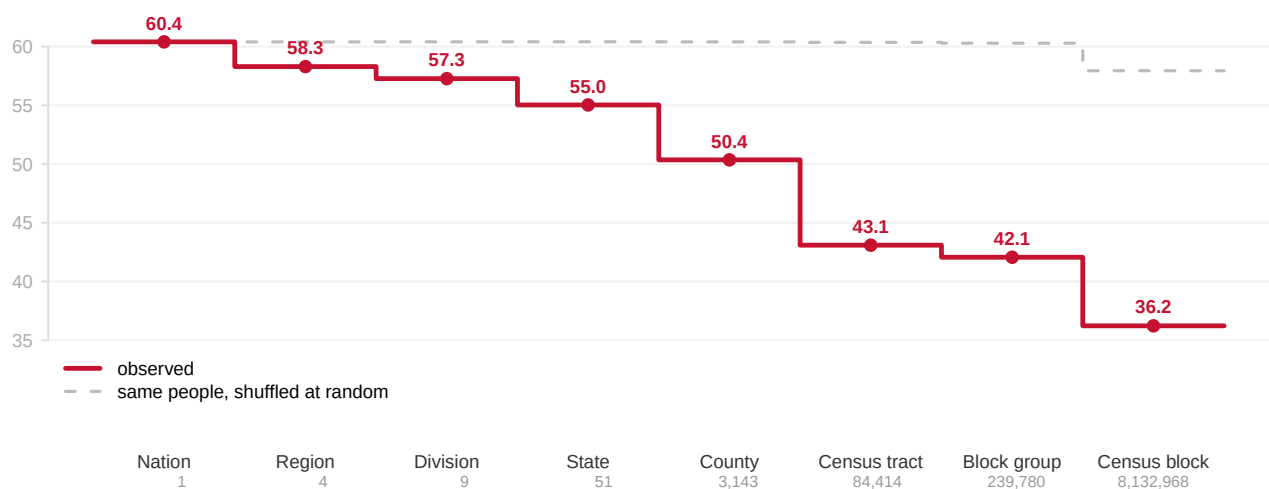
$$E_i = \sum_r p_{ir} \ln \left( \frac{1}{p_{ir}} \right)$$

where  $p_{ir}$  is the share of unit  $i$  in category  $r$ . It is zero when everyone in the unit is in one category. It is at its maximum,  $\ln 8 = 2.079$ , when the eight categories are perfectly even. It is reported below as a percentage of that maximum. So a **diversity score of 100 means the eight categories are equally represented, and a score of 0 means everyone is in the same one**.

The United States as a whole scores **60.4**. The question is what the average American's *surroundings* score, when "surroundings" is allowed to get smaller.

Why entropy and not the dissimilarity index, which is the more famous segregation statistic? Because dissimilarity takes exactly two groups. With eight categories that means either twenty-eight separate numbers, or a decision that one contrast, almost always Black against White, is *the* contrast. That decision is a substantive claim about the country, and the measure should not be making it quietly on our behalf. Entropy takes all eight at once. Its limitations are set out at the end of the chapter, and the first of them is the one this chapter has already spent four pages on: **change the categories and every number below changes**.

### The diversity score of the unit the average American lives in



**Figure 7. Eight rungs, the same 331,449,281 people, one measure.** Each step is the average diversity score of the units at that level, counted by population. It is the score of the unit the average person is in. It falls from 60.4 for the country as a whole to 43.1 for

census tracts and 36.2 for blocks. The dashed line is the control: the identical people shuffled at random into units of the identical sizes.

It stays flat at the national value down to the block-group rung. That means **the fall is not a side effect of using smaller boxes**. Only at the block rung, where units hold a few dozen people, does randomness begin to produce apparent segregation on its own.

Read the two lines together. The matching segregation index is the share of national diversity missing from the average person's unit. It is 0.1663 across counties and 0.2866 across tracts, against a random-shuffle baseline of 0.0007. Essentially all of it is people rather than arithmetic.

That control matters because a companion chapter, *areal-units*, shows the opposite lesson on exactly this kind of ladder. Add Fulton County's blocks upward and the spread of Black share collapses, extremes vanish, and a correlation flips sign. All of it without anybody moving. That lab is about the *measuring instrument*. This figure is what is left when you subtract the instrument, and something is left.

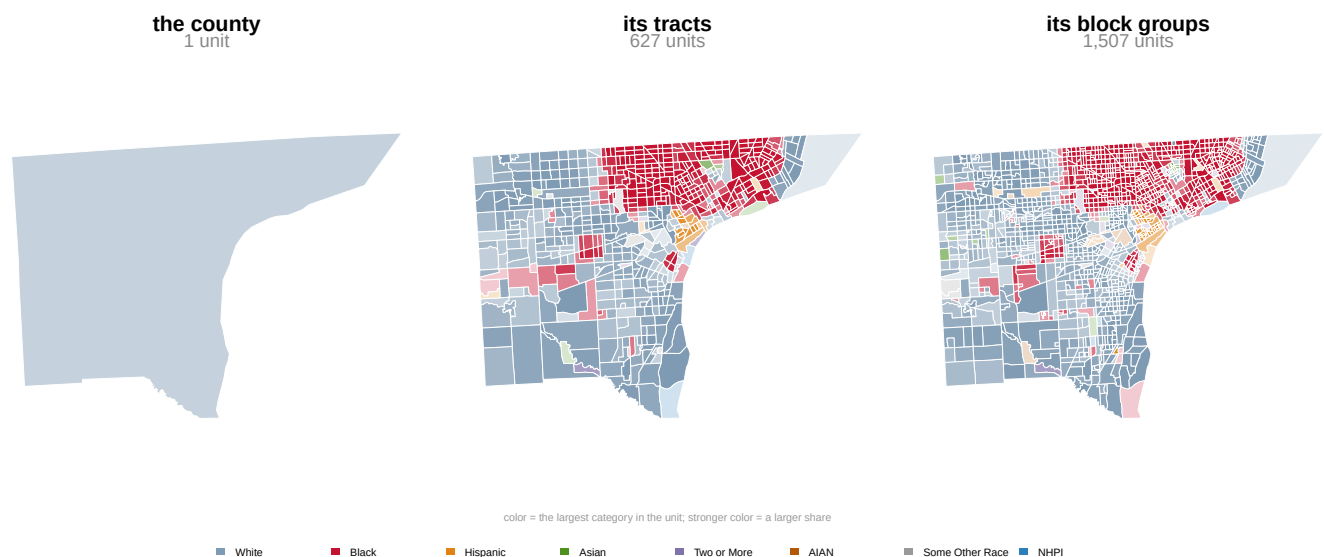
**In plain terms: the United States is 60.4 on this scale and the average American's neighborhood is 43.1.** More concretely still, and without any index at all. **33.9% of Americans live in a census tract where at least four residents in five are in the same category.**

13.1% live in one where at least nine in ten are. Only 17.1% live in a tract whose largest group is under half.

## 08 One county with no majority

Statistics at that scale are still abstractions. So take one county, and take one where the county-level number looks like good news.

**Wayne County, Michigan** holds 1,793,561 people, contains Detroit, and has no majority group at all: it is 47.8% White, 37.3% Black, 6.6% Hispanic and 3.6% Asian. Its diversity score is 57.2, which puts it in the most diverse 6.2% of all 3,143 American counties. If you had to characterize it from the county row alone you would call it one of the more integrated places in the country.



**Figure 8. Wayne County, Michigan, at three scales.** The same 1,793,561 people and the same ground in all three panels; only the size of the unit changes. As one county it is a

single unit whose largest group is white, at 47.8%. Split into 627 tracts it is two counties: 307 tracts where the largest group is White and 279 where it is Black, with a visible boundary between them and very little in between. Split again into 1,507 block groups the boundary sharpens rather than blurring. Color strength is the size of the largest group's share, so the deep colors are the near-uniform units.

The numbers behind the middle panel are the ones to carry away. **The median Wayne County tract is 81.6% a single category.** 201 of its tracts are at least 80% Black, and 125 are at least 80% White. Between them, those 326 tracts hold 51.8% of the county's population. 156 tracts are at least 90% one category, holding 22.9% of the people.

A county in the most diverse seven percent of the United States, in which most residents live in a neighborhood that is at least four-fifths one group. Both sentences describe Wayne County in 2020 and neither is a distortion of the other. **The county row and the tract rows are answers to different questions, and only the tract rows are about anybody's street.**

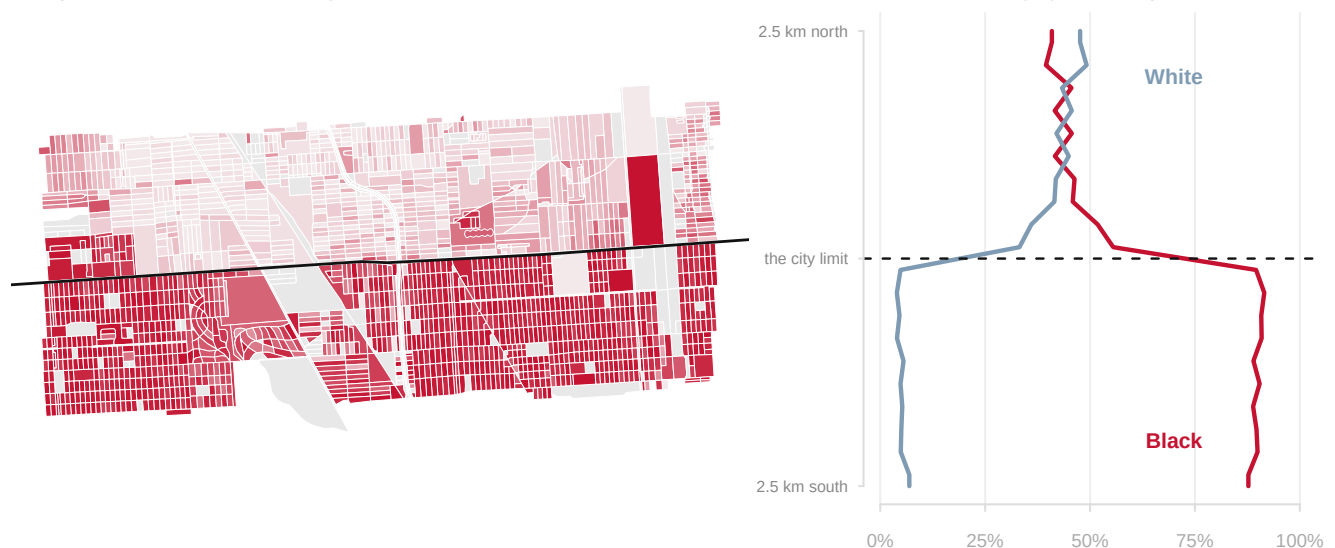
## 09 One street

The boundary visible in the middle panel of Figure 8 is the road this chapter opened on. It was not put there to make a point. It is Eight Mile Road, the northern city limit of Detroit. It fell out of a map coloured by which category is largest, and it is possible to walk across it.

Figure 9 does two things at once. On the left is a 12 km stretch of that line, drawn block by block. It covers 2,234 census blocks holding 110,620 people, in 3 different counties, because along this stretch the city limit is *also* the county line. On the right is the composition of every block within 2.5 km of the boundary along its entire length, by distance from it. The corridor was selected geographically, from the Census Bureau's own published city boundary, before anyone looked at who lived in it.

### blocks along Eight Mile Road

shading = share Black; black line = Detroit city limit



**Figure 9. A line you can stand on.** Left: every census block in a 12 km window of Detroit's northern boundary, shaded by the share of its residents who are non-Hispanic Black. The

heavy line is the city limit as the Census Bureau publishes it. Right: composition against distance from that boundary, computed over 5,445 blocks and 299,535 people along its whole length, in 250-meter bands. The two panels share a vertical axis.

South of the line the corridor is 89.9% Black and 5.0% White, and it is that at every distance. North of it, 45.2% Black and 42.8% White. **Between the last 500 metres inside the city and the first 500 metres outside it, the Black share falls from 90.7% to 53.5%. That is 37.2 points across one road.**

Look at the shape of the profile rather than its levels. It is not a gradient. It is flat, then a step, then flat again. Composition does not fade with distance from Detroit the way something spreading out would. It changes at the boundary and then stops changing.

Whatever produced this pattern was working on the line itself, on which side of a municipal border a house sits. Not on distance from a city centre.

Redlining, restrictive covenants, unequal mortgage access and municipal incorporation all work exactly that way. The redlining chapter follows one of them from a 1939 appraisal form to the ground.

And the administrative point, which was the claim on the first page and now has a figure under it. **The step in the right-hand panel is invisible in every county-level statistic ever published.** The corridor drawn on the left spans 3 counties, for no reason except that the boundary is itself a county boundary. Each of those counties reports a single row in which the step has already been averaged away.

What Figure 9 adds to the opening is the shape. A reader who only had the two county rows would be entitled to imagine a gradient, a city shading into its suburbs over some miles. There is no gradient. There is a step, and the counties are drawn along it.

Figure 7 is why this generalizes past one road. National and state numbers describe places nobody lives; Figure 9 is what that sentence looks like from the pavement. The 3,143 counties of the United States are 3,143 averages; every one of them has boundaries; and there is no reason to expect those boundaries to be demographically quiet.

## 10 The ground did not move

One number from Figure 4 deserves the last word, because it is the exact mirror of the road.

Between 2010 and 2020 the White population of the United States fell by 19,275,992 people. That is a drop of 8.6 percent, the first decline ever recorded in an American census, and it was reported that way everywhere.

Nobody left.

There was no emigration, no epidemic, no annexation. The country grew by 7.4 percent over the decade and the number of people the census reported as White went down by nineteen million. Held against the same two censuses, a second count of the same population shows the White population *rising* by 1.9 percent. That count is equally official, published in the same table, from the same answers on the same forms.

Both numbers are correct. They are not two estimates of one quantity, one of which is mistaken. They are answers to two different questions: *who marked this box and nothing*

else, and *who marked this box at all*. Nothing in the last twenty years has settled which of the two a country is meant to ask.

Set that beside Eight Mile Road and the pair covers the whole failure. At Eight Mile the ground is real, sharp, and sitting still, and the statistic cannot see it. In Figure 4 nothing on the ground moves and the statistic moves anyway, in both directions at once, depending on which of the two counts is read. Nobody is lying in either case. In both, the number and the thing are not the same object, and the gap between them is a property of the instrument rather than of the country.

And the instrument is about to be rebuilt. The revision published on 29 March 2024 (89 FR 22182) folds race and Hispanic origin into a single question. It also adds Middle Eastern or North African, taking the minimum list from 6 categories to 7. Every number in this chapter was computed over 8 boxes that will not exist in 2030.

The diversity score depends on that box list, so it will move. The share of Americans living in a tract that is four-fifths one category depends on it too, so that will move. Eight Mile Road will not move.

Which leaves the two questions this chapter had to ask of every number in it, and that are worth asking of any number about a population. **What is the unit?** The same people score 60.4 as a country and 36.2 as a set of blocks, without anybody crossing a street.

**And what are the boxes?** Between 2010 and 2020 the ground did not move. The question did.

## 11 What this data can and cannot support

**Strengths.** The 2020 Census counts everyone rather than sampling, so there is no sampling error. The file used here is the one the Bureau is required by law to deliver to every state legislature for redistricting. It reaches down to the individual census block, so the same categories can be counted at eight nested scales with no interpolation and no crosswalk. A crosswalk translates one set of areas into another, and none was needed here. Every number in this chapter is a sum of the same records.

**Weaknesses, in the order they would mislead you.**

- **The categories are the finding as much as the data is.** Everything here is computed over eight boxes chosen by OMB and implemented by the Bureau. The diversity score in particular is a function of the category list: merge two categories and it falls, split one and it rises, without a single person moving. Under the 2024 revision it would be computed over 7 categories, and every number in Figure 7 would change.
- **No time series.** For the reasons in Figure 4, this data cannot support a clean statement about how the racial composition of the United States has changed. It can support a statement about how the *measurement* changed, and that is what Figure 4 is.
- **The block rung is the least trustworthy.** The 2020 census applied a disclosure-avoidance system based on differential privacy: state total populations were held invariant and were not altered, but **block-level population counts were**, and the noise is proportionally largest exactly where the units are smallest. The random-shuffle control in Figure 7 shows block-level segregation is still overwhelmingly real, but no single block count in this chapter should be treated as exact.
- **Residents, not voters.** Everything counted here is total population, including children

and non-citizens. Voting-age population is published in the same file and gives different numbers; nothing here is about turnout.

- **One county, one street.** Wayne County was chosen because its county-level number is misleading in an instructive direction, not because it is typical. The national ladder in Figure 7 is what stops the Wayne example doing work it cannot do.
- **Description, not explanation.** Nothing in this chapter establishes why anybody lives where they live. A step function at a municipal boundary is strong evidence that something operated on the boundary; it does not say what.

## 12 What remains genuinely open

- **Whether the 2024 categories will be in use for 2030.** The revision is in force, and the Census Bureau has said it is preparing to implement it in the 2027 American Community Survey and the 2030 Census, with a combined question fielded in the 2026 Census Test. The compliance deadlines have already been extended twice.
- **What happens to forty years of statistics when the categories change.** Every federal series about race is anchored to Directive 15. A person counted as White in 2020 and as Middle Eastern or North African in 2030 has not changed, and no series will say so.
- **Whether the right unit for a claim about segregation is the tract.** It is the conventional choice, and it is a choice. See `areal-units` for what depends on it.

## 13 Sources

### Data

- U.S. Census Bureau, *2020 Census Redistricting Data (P.L. 94-171) Summary File*, legacy format — Tables P1 and P2, for the nation, four regions, nine divisions, 51 states and the District of Columbia, 3,143 counties, 84,414 tracts, 239,780 block groups and 8,132,968 blocks. Fetched 2026-08-11. Keyless; no Census API is used anywhere in this lab. [https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting\\_File--PL\\_94-171/](https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/)
- U.S. Census Bureau, *2010 Census Redistricting Data (P.L. 94-171) Summary File*, national file, for Figure 4 only. [https://www2.census.gov/census\\_2010/redistricting\\_file--pl\\_94-171/National/](https://www2.census.gov/census_2010/redistricting_file--pl_94-171/National/)
- U.S. Census Bureau, *TIGER/Line Shapefiles, 2020* — state, tract, block group, block and place boundaries. <https://www2.census.gov/geo/tiger/TIGER2020/>

### The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`categories.csv`, `counties.csv`, `decades.csv`, `eightmile_profile.csv`, `ladder.csv`, `nation.csv`, `regions.csv`, `standards.csv`, `states.csv`, `tract_largest.csv`, `us_divisions.csv`, `us_maplabels.csv`, `us_rings.csv`, `wayne_bg.csv`, `wayne_bg_rings.csv`, `wayne_block_rings.csv`, `wayne_detroit_rings.csv`, `wayne_outline.csv`, `wayne_tract_rings.csv`, `wayne_tracts.csv`, `wayne_transect.csv`

## Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `ladder.csv` measures the same country at every scale from the nation down. Read `pct_single_group` down the ladder. At which scale does America stop looking mixed and start looking sorted?
- `counties.csv` has both `white_alone` and `white_any`. Take the difference for every county and sort it. Where does that one qualifier do the most work?
- `decades.csv` puts one decade beside the next for every category. Find the category that grew most. Then use `standards.csv` and `categories.csv` to work out how much of that is people changing and how much is the question changing.
- `wayne_tracts.csv` gives each tract a `largest_group` and a `largest_pct`. Count the tracts where one group is almost everybody. Then follow `wayne_transect.csv` across the county line and find where the change happens, and how sharply.

## The standards

- Office of Management and Budget, *Statistical Policy Directive No. 15: Race and Ethnic Standards for Federal Statistics and Administrative Reporting*, 12 May 1977.
- Office of Management and Budget, *Revisions to the Standards for the Classification of Federal Data on Race and Ethnicity*, 62 FR 58782, 30 October 1997.
- Office of Management and Budget, *Revisions to OMB's Statistical Policy Directive No. 15*, 89 FR 22182, 29 March 2024.
- Nicholas Jones, Rachel Marks, Roberto Ramirez and Merarys Rios-Vargas, "2020 Census Illuminates Racial and Ethnic Composition of the Country," U.S. Census Bureau, 12 August 2021 – the source of the quotation in Figure 4.

## The measure

- Theil, H. (1972). *Statistical Decomposition Analysis*. Amsterdam: North-Holland. The information index used here.
- Reardon, S. F. and Firebaugh, G. (2002). "Measures of Multigroup Segregation." *Sociological Methodology* 32(1): 33–67. Why entropy rather than dissimilarity when there are more than two groups.
- Massey, D. S. and Denton, N. A. (1988). "The Dimensions of Residential Segregation." *Social Forces* 67(2): 281–315.

## 14 Related labs

`census-race` takes apart the two questions on the 2020 form and the write-in boxes behind them. `areal-units` shows what changing the unit does to a statistic, and is the control this chapter's Figure 7 needed. `census-geography` explains why the nesting in Figure 8 works at all. `redlining` follows one of the mechanisms that could produce Figure 9. `surnames` and `bisg-check` are about what happens when race has to be *inferred* because nobody recorded it.

## 15 The lab

demographics.Rmd here runs the same sequence as numbered steps with the code exposed, and ends with four things you can change and re-run.

### REBUILD THIS DATA YOURSELF, WITH AN AI ASSISTANT

I want to rebuild a public dataset from its original source, without any starting files. Please do the following and show your work.

THE SOURCE. U.S. Census Bureau, 2020 Census Redistricting Data (P.L. 94-171) Summary File, legacy format, keyless flat files:

– the national roll-up, one small zip:

[https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting\\_File-PL\\_94-171/National/us2020.npl.zip](https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File-PL_94-171/National/us2020.npl.zip)

816,297 bytes, 2,832 geographic records – the nation, its regions and divisions, the states and the metro areas, all in one file

– one zip per state in the same directory, for county-level rows, for example Michigan/mi2020.pl.zip

An honest warning about the second half: all 51 state files (50 states and DC) come to about 1.3 GB compressed. If you only want the shape of the thing, fetch the national file plus one state and say so. Each zip holds pipe-delimited files with no headers: a geographic header, one row per geographic unit, and data segments joining it on the LOGRECNO column. In the first segment, Table P1 (race) starts at field 6 and Table P2 (Hispanic or Latino origin by race) at field 77. Read codes as text. Puerto Rico has a file but is not in the national total.

WHAT TO BUILD.

1. The national composition: total population, Hispanic or Latino, and each non-Hispanic race group, as counts and shares – and each race both ways the Bureau publishes it, "alone" and "alone or in combination". The gap between those two readings is the point.
2. From whichever state files you fetched, a county table of the same categories.

CHECK YOUR WORK. The copies this book used were fetched in August 2026. If your rebuild matches them, you should find:

- a national population of 331,449,281
- 62,080,044 Hispanic or Latino – 18.73%
- non-Hispanic white alone at 57.84%
- 2,832 geographic records in the national file

These are final 2020 census files; they will not change until the 2030 census publishes successors.